Can one reading rule account for the structure of the Standard Model? A compatibility map with registered checks

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Abstract

The structure of the Standard Model can be written with known constructions. Gauge fields are transport matrices on lattice links (Wilson). The electroweak model has a Higgs doublet (Glashow, Weinberg, Salam). The charge pattern of one family is that of Pati–Salam and of Furey's number-operator construction. A single light lattice fermion with exact chiral symmetry comes from the Ginsparg–Wilson relation and Neuberger's overlap operator. This record asks whether one reading rule from the author's framework, Mathematical Materiality (MM), gives these pieces a common meaning. It goes from the data to MM: for each piece, the measured fact, the reading, whether they are compatible, the published construction that already does the same, and what would make the reading incompatible. Selected gauge-sector structures and the charge pattern of one family admit a common interpretation under stated additional hypotheses. The checks concern separate constructions, and their joint chiral gauge realisation remains to be established. No mathematical priority is claimed, and the checks provide no experimental discrimination from the Standard Model. Files labelled "registered predictions" were fingerprinted with SHA-256 before their corresponding calculations; subsequent rechecks are identified separately, and predictions not met are reported. The fingerprints were recorded locally and are not publicly time-stamped.

Version 2.52 of the Mathematical Materiality (MM) series, 15 October 2026. Continues MM 2.40 (the Feynman checkerboard, DOI 10.5281/zenodo.23280168), MM 2.27 (the photon as a phase on the lattice link, DOI 10.5281/zenodo.22667516) and MM 2.38 (anomaly-free charges with a right-handed neutrino, DOI 10.5281/zenodo.23055978). It also reports a counting error in MM 2.40 (section 7).

In the working records the material is in folders named 01_mappa (map, premises and their readings), 02_fermioni, 03_U1 and 04_gruppo_record; working files are in Italian.

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1. What this deposit is, and what it is not

The question. Can a single reading rule give a common meaning to the *structure* of the Standard Model? By structure we mean which forces there are, which charges, how many colours, why the charged weak current acts on one hand, and where masses come from. The work goes from the data to MM, never the other way. For each piece of the Standard Model it states, in order:

- the measured fact;
- the MM reading;
- whether the two are compatible;
- the published construction that already does the same thing;
- what would make the reading incompatible.

The result. Selected gauge-sector structures and the charge pattern of one family admit a common MM interpretation, under the additional hypotheses listed in section 3. The pieces are:

- three colours with eight self-interacting gluon modes;
- the four electroweak bosons with a massless photon and $\rho = 1$ at tree level;
- fractional charges with a neutral neutrino and neutron;

- one light Dirac species on the lattice, from which a Weyl sector is selected by projection; in the U(1) backgrounds tested, its calibrated index agrees with the topological charge.

The checks concern separate constructions; their joint chiral gauge realisation remains to be established. **The parameters are not read:** the weak mixing angle θ_W, the scale v, the masses, the quark and neutrino mixings and the strong CP angle ϑ are taken from the data or left open.

What it is not.

- It is not a derivation of the Standard Model. Every construction used here is known:
- Wilson's lattice gauge theory;
- the Glashow–Weinberg–Salam model;
- the Pati–Salam charge pattern and Furey's number-operator construction;
- the Ginsparg–Wilson relation and Neuberger's overlap operator;
- Kaplan's domain wall.

Section 2 lists them before anything else. **No mathematical priority is claimed.**

- What is specific to MM is only the reading rule that gives these constructions one common meaning, together with the registered checks of where the rule fits the data and where it does not.
- These checks provide no experimental discrimination from the Standard Model.

How the statements are kept apart.

- Published results (section 2).
- Readings: the author's premises translated into physical terms. What the translation adds is marked as the translator's (section 3).
- Numerical checks. Prediction files labelled "registered" were written and fingerprinted (SHA-256, UTC time) before the corresponding calculation. Checks made afterwards are labelled as such. Registered predictions that were not met are reported (section 6).
- Each working round, and this package, was checked by an independent AI system (gpt-6-astra via Codex). Its corrections are applied here (section 7).

2. Prior work, first

topic	published result used here
lattice fermion doubling	Nielsen, Ninomiya, Nucl. Phys. B 185, 20 (1981)†; for discrete time (Floquet): Sun et al., arXiv:1806.09296 (2018), where the theorem still holds; Higashikawa, Nakagawa, Ueda, arXiv:1806.06868 (2018), one Weyl node in a band from a map with winding W = 1
random lattices	Griffin, Kieu, hep-lat/9211022 and hep-lat/9307011: with a gauge-invariant coupling the doublers come back
Ginsparg–Wilson relation and the overlap	Ginsparg, Wilson, Phys. Rev. D 25, 2649 (1982)†; Neuberger, hep-lat/9707022 and hep-lat/9710089 (1998); Hernández, Jansen, Lüscher, hep-lat/9808010 (1999), locality
domain wall fermions	Kaplan, hep-lat/9206013 (1992); Shamir, hep-lat/9303005 (1993); Golterman, Jansen, Petcher, Vink, hep-lat/9309015 and hep-lat/9312005 (1993)
chiral gauge theories on the lattice	Lüscher, hep-th/0102028 (2001): abelian chiral gauge theories constructed non-perturbatively, non-abelian ones to all orders of perturbation theory; Kaplan, Les Houches lectures, arXiv:0912.2560 (2009): no practical way is known to regulate general non-abelian chiral gauge theories on the lattice; Kadoh, Kikukawa, arXiv:0709.3658 (2008): a gauge-invariant non-perturbative construction of the Glashow–Weinberg–Salam model, covering all SU(2) topological sectors with vanishing U(1) magnetic flux
lattice gauge fields	Wilson, Phys. Rev. D 10, 2445 (1974)†: transport matrices on links, plaquette action
electroweak model	Glashow (1961), Weinberg (1967), Salam (1968)†; PDG 2025, review 10 "Electroweak model and constraints on new physics"
charges from a number operator	Furey, arXiv:1603.04078 (2016) and arXiv:1405.4601 (2014)

the Pati–Salam charge pattern	Pati, Salam, Phys. Rev. D 10, 275 (1974)†, the lepton as a fourth colour; SO(10) unification (PDG 2025, review "Grand Unified Theories")
anomaly-free charges with ν_R	MM 2.38 (DOI 10.5281/zenodo.23055978): with a right-handed neutrino and Dirac masses the anomalies leave the hypercharge of the quark doublet free
quantum cellular automata	D'Ariano, Perinotti, arXiv:1306.1934 (2014); Bisio, D'Ariano, Tosini, arXiv:1212.2839 (2015); MM 2.40
Higgs self-coupling at the Planck scale	Shaposhnikov, Wetterich, arXiv:0912.0208 (2010): $m_H \approx 126$ GeV from $\lambda(M_P) \approx 0$ under their ultraviolet assumptions; Buttazzo et al. (2013), as used in MM 2.38
mass hierarchy from a fifth direction	Arkani-Hamed, Schmaltz, hep-ph/9903417 (1999)
winding of a product of maps	a product identity for the winding, derived in the working notes and reproduced in section 4.7; in the literature it is associated with the name Polyakov–Wiegmann (not checked at the source here)

† Standard references cited from the reviews listed, not re-read at the source for this deposit.

Data. PDG 2025 (S. Navas et al., Phys. Rev. D 110, 030001 (2024) and 2025 update): reviews on the electroweak model, Higgs boson, number of light neutrino types, muon decay parameters, CKM matrix, neutrino mixing, conservation laws; summary tables of mesons and baryons. Bagdonaite et al., arXiv:1311.3438 (2013), $\Delta\mu/\mu$ from methanol at $z = 0.89$. Burns, Keus, Sher, Tait, arXiv:2402.08626 (2024), and Yoo, Scherrer, astro-ph/0211545 (2003), v at nucleosynthesis. Uzan, arXiv:1009.5514 (2011), review of varying constants.

3. The reading rule, in physical terms

The author's premises (ten of them on 15 October 2026, with earlier ones from the MM book and from 27 September 2026) are stated here in physical terms. The exact wording, in Italian, appears only in the original registrations (`documenti/registrations_original/`). An informal account of the picture behind them, with its terms defined, is in `WHERE_THIS_COMES_FROM.md`; nothing in the checks depends on it.

The rule in one sentence. Particles and forces are not different objects but different readings of the same structure. Each reading assigns:

- to each link of the lattice, a transport of the readings involved: one phase for light, two units for the weak force, three readings for colour;
- to each fermion, the number of readings it shares (its charge) and the hand it is read with (its chirality);
- to the Higgs field, the projection that links the two hands (the mass).

Its parts. In the third column, what the translation adds to the author's words.

	reading	added by the translation		
R1	**Three internal readings.** The internal space of a fermion is spanned by three binary readings (taken or not): $2^3 = 8$ states. The colour is which readings are involved; a transport along a link may mix them, a unitary 3×3 matrix.	That the three readings are equivalent and may be mixed by any unitary matrix. That the transport preserves a chosen complex volume form on C^3, which imposes $\det U = 1$, so SU(3) rather than U(3).		
R2	**Charge from shared readings** (rule R-Q). \	Q\	= N_shared/3, with N_shared ∈ {1, 2} for colour triplets (quarks) and {0, 3} for singlets (leptons); conjugate states carry opposite charges.	The split between triplets and singlets is an added rule that contains the quark –lepton distinction.
R3	**Three families.** The same triple of readings, seen from the other side, gives the three families.	That the number of colours equals the number of families. Only the number is read, not the masses.		
R4	**Light and the weak force.** Light is a phase on the	The identification of the "slightly smaller unit"		

	link (MM 2.27). The weak force is the same transport with two units mixed on the link, one full and one slightly smaller: a U(2) = [SU(2) × U(1)]/Z_2 matrix.	with the second state of the doublet.
R5	**Mass and the Higgs.** The mass of a fermion is the projection that links its two hands; the Higgs field is where this projection happens, with one scale v. The Higgs mass comes from the self-coupling of that field. The low-temperature electroweak scale is a fixed structural parameter.	That "self-resonance" is the quartic coupling λ; the reading of "history" as structure (L8-a) rather than as elapsed time (L8-b).
R6	**One hand as an act of reading.** For charged fermions, the left- and right-handed components form a Dirac field after electroweak symmetry breaking. The charged weak current acts on the left-handed component, light on both. The reading places the difference in the projector rather than in the field. The lattice twin of the light species is pushed to the scale of the cell when the reading level lies above the bottom of the "hill" of the lattice spectrum and below its first ridge.	The two components carry different electroweak representations, and the Yukawa coupling remains part of the construction: the projector alone does not assign them. The window $0 < m_0 < 2r$ is a condition of the construction, not a consequence of the premises. Reading the author's "the top of the hill is passed" as "the bottom of the valley is passed" is the translator's new interpretation of the metaphor. The Dirac or Majorana nature of neutrinos is left open.

4. The map, piece by piece

4.1 Charges of one family (registered P2a–P2c; `codice/03_U1`)

- **Data.** Charges 0, ±1/3, ±2/3, ±1; neutron and hydrogen atom neutral.
- **Reading.** R1 + R2, rule R-Q.
- **Check.** In the anomaly-free family with ν_R and Dirac masses (MM 2.38), the charges depend on the free hypercharge Y_Q of the quark doublet: q_u = Y_Q + 1/2, q_d = Y_Q − 1/2, q_ν = 1/2 − 3Y_Q, q_e = −1/2 − 3Y_Q. The four anomalies vanish for every Y_Q. Rule R-Q admits exactly Y_Q ∈ {−1/6, +1/6} (search over rationals with denominator up to 60 in [−3, 3], and an exact solution).
- Y_Q = +1/6 is the Standard Model: neutrino 0, electron −1, u 2/3, d −1/3, neutron 0, hydrogen 0.
- Y_Q = −1/6 is the same content after Q → −Q and exchanging the two members of every doublet (u ↔ d, ν ↔ e, with their right-handed partners, the Higgs with its conjugate and the Yukawa couplings), colour unchanged. This was checked afterwards; the registered form of the relabelling (Q → −Q with 3 ↔ $\bar{3}$) was wrong.
- Without ν_R the rule adds nothing: the anomalies already fix Y_Q = 1/6, and the rule admits it.
- **Compatible: yes.** With ν_R the rule fixes the charge where the anomalies alone leave it free.
- **Place already taken.** Pati–Salam and SO(10) also fix the charge assignments with ν_R, and Furey (2016) obtains the charge pattern from a number operator on the same 8-state space. R-Q is not a derivation of those models; it is an added rule that contains the quark–lepton distinction.
- **Not read.** The value of the fine-structure constant, the Yukawa couplings.

4.2 The algebra of the three readings (registered T-A; `codice/01_premesse`)

The 8 states of R1 carry fermionic operators ε_i (i = 1, 2, 3) of a three-mode Fock space (the exterior algebra of C^3), as in phase 2.29 of the MM 2.27 deposit and in Furey (2016).

- A1: canonical anticommutators, exact (error 0).
- A2: the number operator has spectrum {0, 1, 1, 1, 2, 2, 2, 3}, so N/3 gives 0, 1/3 (×3), 2/3 (×3), 1. In the archived operator convention Σ ε_i†ε_i counts the empty readings (3 − Σb) and N_shared = Σ ε_i ε_i†; the sector labels N = 1 and N = 2 in the log are swapped with respect to the operator (reviewer). Spectrum and Casimir values do not change.
- A3: the nine ε_i†ε_j commute with N; their traceless part generates su(3), with Casimir 4/3 on the sectors with one and two readings (two conjugate triplets) and 0 on the other two.
- A4: the spin generators of phase 2.29, S_i = −(i/2) C_j C_k, do not commute with N (‖[S_i, N]‖ = 1). Stated conditionally: these spin generators cannot act on the internal index with this charge. A representation is needed in which spin and charges commute, usually with separate factors.
- A5: the fifteen products C_iC_j, K_iK_j, C_iK_j conserve the parity of N and generate an algebra of dimension 15, irreducible on the even and odd sectors (1 + 3 and 3 + 1). Correction (reviewer, rechecked): these raw products generate so(3,3) ≅ sl(4,R), a non-compact form. With the Hermitian generators (C_i, iK_i) the algebra is the compact su(4), of dimension 15, with 4 → 3 + 1 (`ricontrollo_ta_dopo_astra.py`). This is the Pati–Salam pattern (the lepton as a fourth colour), not the full Pati–Salam model: the weak factors, the identification of the physical generators and the dynamics are missing.

4.3 Three colours, three families (registered T-B; data only, low weight)

- **Data.** N_ν = 2.9963 ± 0.0074 light neutrinos coupled to the Z (0.5σ from 3). R_ℓ = Γ(hadrons)/Γ(leptons) at the Z is roughly proportional to the number of colours: 2 or 4 colours would move it by about 33 %.
- **Reading.** R3: the same triple read from two sides.
- **Compatible: yes, with low weight, as registered.** The number three appears everywhere in MM (three directions, eight states = 2^3, rank 3 of the book's decomposition, which holds when the prime supports are disjoint), so a rule giving 3 is not surprising. The figure 1/3 in the registration is an example with an assumed distribution, not a significance. N_ν counts light active neutrinos coupled to the Z, not all possible families.

4.4 Gluons (registered G1–G3; `codice/04_gruppo_record/gluoni.py`)

- **Data.** Eight massless gluons that interact with each other; no ninth long-range force coupled to baryon number.
- **Reading.** The photon method of MM 2.27 (a phase on the link) with the transport of the three readings (R1): a 3×3 matrix U = exp(iA) on each link, plaquette action S = Σ_p Re Tr(1 − U_p), static fields on a periodic 6^3 lattice.
- **Check.**
- G1: at each of the three tested wavevectors ((1,0,0), (1,1,0), (1,2,3)) the static quadratic form has 16 complex transverse amplitudes for su(3), that is 8 × 2. (The program prints half the rank of the real sine/cosine form, 32.)
- G3: with u(3) instead of su(3), 18: the ninth appears only without det = 1.
- G2: a cubic term in $S - S_2$ for a generic non-commuting su(3) configuration: $c_3 = -39.0$ from a fit $c_3 + c_4\varepsilon$ at ε down to 1e-4 (recheck afterwards). For commuting fields the cubic term vanishes, while the compact plaquette action keeps higher even powers. The registered criterion (ratio constant within 5 % between ε = 1e-2 and 1e-3) was not met, because at ε = 1e-2 the quartic term weighed as much as the cubic one.
- **Compatible: yes.** These are transverse amplitudes of the static quadratic form, not verified counts of propagating polarisations; c_3 is a coefficient of that configuration. det = 1 is the added hypothesis of R1. The central U(1) of U(3) is not automatically baryon number.
- **Place already taken.** Wilson (1974): with matrices on links, Yang–Mills follows by construction. The MM part is only the meaning of the link (three readings) and the reason offered for det = 1. Confinement and the running of α_s are not redone here.

4.5 The electroweak bosons: two units on the link (registered E1–E4; `codice/04_gruppo_record/debole_due_unita.py`)

- **Data.** W^+, W^-, Z, γ; the W have charge ±e; the triple and quartic gauge couplings measured at LEP 2 are those of a non-abelian theory; M_W = 80.369 GeV, M_Z = 91.188 GeV, massless photon; ρ = 1 at lowest order.
- **Reading.** R4 + R5: a U(2) transport on the link, and a Higgs doublet frozen at its vacuum (0, v/√2) in unitary gauge.
- **Check.**
- E1: at each of the three tested wavevectors, 8 complex transverse amplitudes (4 bosons × 2).
- E2. The registered cubic-term statement is supported by the generic-field calculation ($c_3 = -29.4$) and by the absence of a cubic term for commuting fields. For W3 and B only, $(S - S_2)/\varepsilon^3$ goes to zero like ε (−0.77, −0.23, −0.06), and the absence is also analytic: with commuting fields the plaquette is exp(iεH_p), with H_p Hermitian and independent of ε, and the real part of its trace contains only even powers of ε. The original program returned "not met" under a fit criterion that is not in the registration: an intercept of 0.0144 against a tolerance of 1e-3. Loss of numerical precision in the subtraction is a possible explanation of the residual; no higher-precision comparison is supplied here.
- E3. The registration gives the physical masses squared: 0, $g^2v^2/4$ (×2) and $(g^2 + g'^2)v^2/4$. The program built the matrix K of the quadratic form c^T K c, whose eigenvalues are 0, 3201.7 (×2) and 4164.7 GeV², and compared them directly with the predicted physical values. For canonically normalised real fields $M^2_{phys} = 2K$, so the original automatic comparison omitted a factor of two and returned "not met". The original program also printed ρ = 2, from a formula that put the predicted physical M_W^2 in the numerator and the computed eigenvalue K_Z in the denominator (a common rescaling of both masses would leave ρ unchanged). The subsequent calculation uses consistent normalisations: it recovers the registered values, 0, 6403.44 (×2) and 8329.35 GeV², and ρ = 1.000000000000. This is a tree-level identity; the twelve digits describe the numerical check, not an experimental precision. The photon direction W3/B = tan θ and the W charge +1 in units of the photon coupling were checked in the original program. g = 0.65 is chosen, which gives M_W = 80.022 GeV and M_Z = 91.265 GeV.
- E4: M_W/M_Z = cos θ with the MS-bar value $\sin^2\theta = 0.23122$ gives 0.8768 against 0.8814 measured (−0.52 %). This was registered in advance as the known MS-bar versus on-shell difference, not a test of MM.
- **Compatible: yes, for selected bosonic identities.** With a U(2) link representation the checks reproduce:
- four bosons;
- a cubic self-interaction for non-commuting fields;
- a massless photon;

- massive W and Z with ρ = 1 at tree level, and W charge ±1.

This is not yet an assembled electroweak model. The normalisation of the two gauge kinetic terms, which fixes g and g′ independently, and the action on the fermion representations are not constructed here.

- **Place already taken.** The Glashow–Weinberg–Salam model on a lattice: with 2×2 matrices on links and a doublet, all of this follows by construction. The MM part is the meaning of the two states.
- **Not read.** θ_W (g′/g is taken from the data), v, and which hand the weak force acts on. The ratio of the two units might one day fix g′/g at some scale; no such rule was written in advance, and none is sought afterwards.

4.6 One light hand (registered O1–O5, H1–H2; `codice/02_fermioni/overlap_240.py`, `o5_h2_rapido.py`, `collina_h1.py`)

- **Data.**
- The charged weak current acts on left-handed fermions. PDG 2025, review 57 (muon decay parameters), table 57.1, 90 % limits: |g^V_LL| > 0.960, |g^V_RR| < 0.017, |g^V_LR| < 0.023, |g^V_RL| < 0.105.
- Within the local, derivative-free, lepton-number-conserving four-fermion framework reviewed there, muon-decay and inverse-muon-decay measurements, together with the helicity −1 of ν_μ in pion decay, support the V−A leptonic charged interaction within experimental errors. These limits concern specific operators and low-energy processes.
- No light mirror fermions with the same weak charges are observed. For charged Standard Model fermions, the Z has both left- and right-handed couplings; an electroweak-singlet ν_R has no such coupling.
- **The obstacle.** Under the assumptions of the Nielsen–Ninomiya theorem, a lattice formulation has fermion doubling; the overlap construction uses the Ginsparg–Wilson form of chiral symmetry. Section 4.7 shows what the doubling means for the MM 2.40 checkerboard.
- **Reading.** R6, implemented as a standard overlap operator built with the matrices and parameters of MM 2.40. The Dirac matrices come from the checkerboard, with Wilson parameter r = 0.15 and level m_0 = 0.2 (the mass of the 2.40 domain wall). Time is treated as a fourth Euclidean direction. In formulas: D_W(k) = i Σ γ_μ sin k_μ + r Σ (1 − cos k_μ), A = m_0 − D_W, D = 1 − A(A†A)^(−1/2). The standard set (r = 1, m_0 = 1) is run alongside. **It is not shown** that this Euclidean operator is the limit of the 2.40 tick automaton.
- **Check.**
- O1: the Ginsparg–Wilson relation γ5D + Dγ5 = Dγ5D holds to 9e-16 for both sets.
- O2: in the free field D has a single zero, at k = 0, and D = 2 at the 15 corners. Outside |k| < 0.3 the smallest singular value on a 16^4 grid is 1.056 (2.40) and 0.390 (standard).
- O3: with $\hat{\gamma 5} = \gamma 5(1 - D)$, the Weyl operator $P_+ D \hat{P}_-$ has one cone at k = 0 (slope $1/m_0$ = 5 for 2.40) and is gapped elsewhere.
- O4, locality. The kernel norm f(r) at Manhattan distance r decays with effective fitted slopes 0.4054 over distances 4–14 (2.40) and 0.8049 over distances 2–8 (standard). Both calculations used 48^4 sites instead of the registered 32^4. The registered range for 2.40 was 0.1–0.4, so O4 was not met for 2.40 by a small margin. For the standard set the numerical criterion is met, on the modified lattice.

The fitted slopes are not asymptotic decay rates. An axial complex-momentum singularity of the 2.40 kernel lies at k = (π + iκ, 0, 0, 0) with κ ≈ 0.101. A diagonal one lies at k = (π + iη, iη, iη, iη) with η ≈ 0.050 (reviewer; both rechecked). The asymptotic decay rate in the Manhattan norm used in O4 is therefore not established here. The operator is local with exponential tails, not ultralocal. **The earlier working phrase "the twin fades over 2.5 cells" is withdrawn.**

- O5, the index with the light field. The lattice is 6^4, with constant U(1) fluxes ('t Hooft). The calibrated index equals the topological charge Q for Q = 0, 1, 2, −1, 0, both for 2.40 and for the standard set, and the gap of H_W stays open (≥ 0.0999 and ≥ 0.74). For 2.40 the registered expectation was a mismatch at L = 6, because the flux per plaquette (0.17) exceeds the margin 2r − m_0 = 0.1. The anticipated mismatch was not observed in the five tested backgrounds; no general conclusion on reliability follows.
- Index convention. With H_W = γ5(m_0 − D_W), the identity $n_+ - n_- = \tfrac{1}{2}$ Tr sign(H_W) holds. `overlap_240.py` counts exact zero modes as $n_- - n_+$; after the first background, `o5_h2_rapido.py` uses the spectral asymmetry. The two methods agree on the first background, Q = 0, where the sign is not tested. The sign convention is calibrated separately at Q = 1, as registered, so the index is not an independent prediction. The fast method does not exclude extra zero-mode pairs of opposite chirality.
- H1, the "hill": the number of light species and their net chirality (the coefficient of the index after the Weyl projection) as functions of m_0/r = −0.5, 1, 3, 5, 7, 9 are 0/0, 1/+1, 5/−3, 11/+3, 15/−1, 16/0, for r = 1 and r = 0.15.
- H2: with Q = 1 the index is 1, −3, 3, −1, 0 for m_0 = 1, 3, 5, 7, 9.
- **Compatible: yes, in a window.** The level must lie above the bottom of the valley and below the first ridge (0 < m_0 < 2r); the 2.40 values (0.2 < 0.3) are inside. Above the top (m_0 > 8r) all the twins return and the net chirality cancels. In the Ginsparg–Wilson form the projector of one field depends on the operator itself, $\hat{P} = (1 - \gamma 5(1 - D))/2$, so the difference between the hands sits in the reading operator. This is R6 in a known object, a

compatibility and not a derivation.

- **Precisely what is shown.** The overlap operator has one light Dirac species, with both chiralities; a Weyl sector is selected by projection. These checks do not construct the fermion measure of a chiral gauge theory, not even for U(1), and everything here is abelian. Non-perturbative lattice constructions of the electroweak model exist with stated restrictions (Kadoh–Kikukawa 2008); implementing such a construction is outside this deposit.
- **Place already taken.** Neuberger (1998), Hernández–Jansen–Lüscher (1999), Kaplan (1992), Shamir (1993). The dependence of the number of species and of the index on m_0 is known in the overlap literature.
- **A side check on pion decay** (`pione_elicita_v2.py`). For massless neutrinos and lepton-universal V/A couplings, the tree-level ratio $\Gamma(\pi \to e\nu)/\Gamma(\pi \to \mu\nu)$ is $1.2833 \cdot 10^{-4}$, independent of the chiral mixture. The measured $1.230 \cdot 10^{-4}$ is consistent with this helicity suppression; Standard Model radiative corrections account for a reduction of roughly four percent, not computed here. A lepton-universal non-derivative pion–lepton coupling without an additional lepton-mass factor would give 5.487. The ratio does not select a hand. **An earlier working note (`pione_elicita.py`) read the ratio as evidence for one hand; that reading is corrected here.** The evidence for the left-handed charged current is the muon-decay data above.
- **Would be incompatible if:**
- an established right-handed charged-current contribution were found that is incompatible with the specified chiral interaction;
- a light mirror sector with the stated charges were found;
- the non-locality of the reading operator were observable at long range.

4.7 The obstacle on the 2.40 checkerboard: winding (registered T1–T6; `codice/02_fermioni/tappa1_*.py`)

The 2.40 checkerboard moves each tick by one step along x, y and z in turn. For the Weyl sector this is a map $U: T^3 \to SU(2)$. Its three-dimensional winding number is $W = -(1/24\pi^2)\int \varepsilon_{ijk}\, \mathrm{Tr}[R_i R_j R_k]\, d^3k$, with $R_i = U^\dagger \partial_i U$. For SU(2), when $\pm I$ are regular values, W equals the sum of node chiralities defined from the local effective Hamiltonian at quasi-energy 0, and separately at π, with a consistent overall sign convention. The registered definition (the sign of the Jacobian of the vector part u of $U = u_0 I - i\, u\cdot\sigma$) agrees with this at quasi-energy 0 but has the opposite sign at π, where the local effective field is $-u$ (reviewer).

- T2: $W = 0$ for MM 2.40 ($-2e{-}16$ on 48^3 and 96^3 grids).
- T3: $W = 0$ for 200 random automata made of single-direction steps in SU(2) (largest $|W| = 1.2e{-}14$). This is also proved: each step is a loop in SU(2), and $\pi_1(SU(2)) = 0$.
- T3b: the Weyl automaton of D'Ariano–Perinotti is exactly a product of three single-direction steps, $A\pm =$ e^{−iσ_z q_z} e^{±iσ_y q_y} e^{−iσ_x q_x}, so $W = 0$.
- T4 (control): the model of Higashikawa et al. gives $W = +1.00000$, with the sign convention that reproduces their dispersion; $W = 1$ is met. The unrestricted registered statement "a single gapless point" is not met: uniqueness holds at quasi-energy zero, while degeneracies also occur at π on the zone faces.
- T5a: partial steps (one spin component moves, the other waits), three steps: $W = 0$ in 64 of 64 cases.
- T5b and T5c were not met. The registered expectation was some $W \neq 0$ with 4–8 partial steps, but all 236 had $W = 0$. On the doubled cell, Higashikawa's model gives $+1 - 1 = 0$, not 2. Both are explained afterwards by an identity: $\mathrm{Tr}[((gh)^{-1}d(gh))^3] = \mathrm{Tr}(g^{-1}dg)^3 + \mathrm{Tr}(h^{-1}dh)^3 - 3\, d\,\mathrm{Tr}(g^{-1}dg \wedge dh\, h^{-1})$, so $W(gh) = W(g) + W(h)$ for single-valued regular maps on the torus.
- **Consequence:** a finite product of steps, each periodic on the same torus and depending on one component of k, with constant coins, has $W = 0$ in U(n). The conditions matter: the coins must be constant (or have known W), and each factor must be periodic, not only the product. Higashikawa's half steps are not single-valued on the cell.
- **What it means.** In the checkerboard with separate steps, the Weyl sector does not realise one isolated hand in the bulk. To obtain a chiral effective sector one must state which further conditions or sectors enter; section 4.6 does so with the reading level. The earlier working phrase "the opposite hand can only be moved" is withdrawn: corners, bands and walls are different mechanisms with their own assumptions.
- T1c: the registration expected, with the Wilson term $r = 0.15$, one light species and 15 heavy species with gap ≥ 0.1. The single light species is confirmed, and the other valleys have a gap ≥ 0.160 (grid 40^3). The species count is 64 valleys, not 16 (section 7), and the original minimum search was incomplete.
- T6, dispersion. The registration concerned the leading order of the group-velocity deviation; the program computed the relative energy deviation. The leading orders are supported, and the coefficients were corrected analytically afterwards:
- one tick (2.40 without the palindrome, and D'Ariano–Perinotti): relative energy shift linear in κ, with coefficient $n_x n_y n_z$ and opposite sign for the two helicities;
- the palindrome and the 2.27 photon: only quadratic shifts. For the photon, with $k = \kappa n$, the relative energy deviation is $-(\kappa^2/24)\, \Sigma\, n_i^4$ and the radial group-velocity deviation $-(\kappa^2/8)\, \Sigma\, n_i^4$; the registration had quoted 1/24 for the velocity (checked afterwards).

The registered phenomenological consequences (a linear term at the Planck scale compared with Lorentz-violation limits) are not verified in this deposit: the comparison would need the operator, sector and physical step fixed in advance.

4.8 The Higgs sector (readings compared with data; no new code)

- **Couplings and mass.** The measured Higgs couplings are consistent with the Standard Model pattern (PDG 2025): tree-level fermion couplings scale as m_f/v and W/Z couplings as m_V²/v; photon and gluon couplings are loop-induced. R5 reads the fermion mass and its Higgs coupling as the same projection seen twice. **Compatible.** Two limits apply:
- "one scale for all masses" holds in the electroweak sector only, since the nucleon mass comes from QCD;
- "no right hand, no mass" holds for Dirac masses, and Majorana terms exist.
- **The Higgs mass as self-coupling.** In the Standard Model $m_H^2 = 2\lambda v^2$ ($\lambda = 0.129$ at first order). The trilinear coupling is measured at $-1.2 < \kappa_3 < 7.2$ (ATLAS, 95 %), against 1 in the Standard Model. **Compatible by structure;** today's data do not discriminate. A κ_3 different from 1 would not by itself contradict the reading.
- **$\lambda = 0$ at the Planck scale** (an additional condition, not contained in the premises). From MM 2.38: $\lambda(M_P) = -0.0090 \pm 0.0052$ with the pole top mass (1.7σ from 0) or -0.0100 ± 0.0031 with the direct top mass (3.2σ). The place is taken: Shaposhnikov–Wetterich (2010).
- **Numerology declared in advance and not used.** $m_H/v = 0.5085$ against 1/2; $\lambda = 0.1293$ against 1/8. As an exact first-order relation, $m_H = v/2$ is excluded (123.11 against 125.20 ± 0.11 GeV).
- **v and cosmic history.**
- Methanol at $z = 0.89$ (7.5 Gyr ago): $\Delta\mu/\mu = (-1.0 \pm 0.8_{stat} \pm 1.0_{sys}) \cdot 10^{-7}$, with $\mu = m_p/m_e$ (Bagdonaite et al. 2013).
- Nucleosynthesis: $|\delta v/v| \lesssim 1$ %, a model-dependent bound (Burns et al. 2024).
- Turning the μ limit into a limit on v needs $\Delta\ln\mu = \Delta\ln m_p - \Delta\ln y_e - \Delta\ln v$, with m_p tied to QCD. Bounds on a cosmological variation of v depend on the dimensionless quantities compared and on assumptions about the Yukawa couplings and QCD; they refer to the low-temperature vacuum value, not to the thermal history of the electroweak transition.

The reading L8-a takes the low-temperature electroweak scale as a fixed structural parameter. It is **compatible with these bounds, but not singled out uniquely.** The value $v/M_P \approx 2 \cdot 10^{-17}$ (M_P the non-reduced Planck mass) is not read.

4.9 SU(2) from the records (registered, working result only; no code in this deposit)

A different route was tried for the weak group: the holonomy group of the transport around closed histories of records, on the record benches of the 2.34 working programme. The registered expectations were not met.

- With 4 registers: semisimple algebras of dimension 6 (so(4)) or 3 (so(3)), depending on a parameter.
- With 3 registers: abelian (dimension 1) or trivial.
- With 5 registers: undetermined.

Algebra, group and force must be kept apart ($so(4) \cong su(2) \oplus su(2)$; $SO(4) \cong [SU(2) \times SU(2)]/Z_2$; SO(3) is not SU(2)). **No stable SU(2) is read from the records today.** The programs depend on unpublished working files of the 2.34 programme and are not included; the registrations and the outcome note are (`documenti/`).

5. What is read, what is not, and what would contradict

piece	read (structure)	not read or not constructed	observation incompatible with the stated rule or realisation
charges	0, ±1/3, ±2/3, ±1; neutral ν and neutron; $Y_Q = \pm 1/6$ with ν_R (4.1)	α, Yukawa couplings	charges or colour assignments violating R-Q within its stated scope; a non-zero neutrino or neutron charge under the assumptions of section 4.1
colour	SU(3) transport, 8 self-interacting modes of the static form (4.2, 4.4)	confinement, running of α_s (not redone)	evidence that the colour transport requires an additional central U(1) generator coupled to the same internal states
colours and families	3 and 3 (4.3, low weight)	why the families differ in mass	an additional fermion family, or a colour multiplicity different from three (the Z-width check tests only light active neutrinos)
electroweak bosons	γ, W±, Z; cubic self-interaction; ρ = 1 at tree level; W charge ±1 (4.5)	θ_W, v, M_W, M_Z; the gauge kinetic normalisation; the action on fermions	evidence that the gauge-boson masses cannot be described by the single-doublet realisation used here (additional scalar representations

			need not imply $\rho \neq 1$)
one hand	one light Dirac species, a Weyl sector by projection, calibrated index = Q in the tested U(1) backgrounds, in the window $0 < m_0 < 2r$ (4.6)	an anomaly-free chiral fermion measure and the full electroweak coupling	an established right-handed charged-current contribution incompatible with the specified chiral interaction; a light mirror sector with the stated charges; observable long-range non-locality
Higgs	tree-level fermion couplings m_f/v and W/Z couplings m_V^2/v, photon and gluon couplings loop-induced; one electroweak scale; m_H from self-coupling (4.8)	v, v/M_P, the hierarchy, κ_3	Higgs-coupling measurements incompatible with the single-doublet pattern used here, including its radiative corrections
λ at the Planck scale (additional condition)	$\lambda(M_P) \approx 0$ (4.8)	—	$\lambda(M_P)$ clearly different from 0 with a better top mass
v in time	a fixed low-temperature electroweak scale (4.8)	the value of v	an established variation of a specified dimensionless ratio involving v, incompatible with the fixed-scale interpretation under the stated Yukawa and QCD assumptions
masses	—	all fermion masses (taken from data)	—
mixings	—	CKM, PMNS, CP phases	—
strong CP	—	$\bar{\theta}$, constrained indirectly by electric-dipole-moment limits to order 10^{-10}	—
neutrinos	—	the mass mechanism: the premises do not select it	— (observing neutrinoless double-beta decay would establish lepton-number violation; not observing it would not by itself establish Dirac neutrinos)
weak group from records	— (bench-dependent, 4.9)	SU(2) from the records	—

6. Registered predictions and outcomes

Full list, with abridged English translations of the registrations, in `documenti/REGISTERED_TESTS.md`; the original Italian files and the registration logs are in `documenti/registrations_original/`.

block	registered	met	not met as registered
T-A three readings	A1–A5	A1–A4; A5 in dimension and irreducibility	A5 identification (so(3,3) for the raw products, corrected)
T-B colours and families	one comparison	yes (low weight, as registered)	—
T-D wall states of 2.40	D1, D2	D2	D1 (see section 7)
Tappa 1 winding	T1a–T6	T1a (in substance), T2, T3, T3b, T4 (W = 1), T5a, T6 (leading orders)	T1b (64 valleys, not 16); T1c species count (the single light species holds); T4 unrestricted single gapless point; T5b; T5c
Tappa 2 charges	P2a–P2c	P2a, P2c	P2b (form of the relabelling)
Tappa 3 records	G1–G6	G2	G1, G3, G4, G5 and G6 (n = 3); n = 5 undetermined
gluons	G1–G3	G1, G3	G2 (criterion; cubic term present in the recheck)

electroweak	E1–E4	E1 (at the three tested wavevectors), E2, E3 (recovered with consistent normalisation), E4 (as registered, not a test)	— (the original program returned "not met" for E2, under an unregistered fit criterion, and for E3, through a normalisation error in its automatic comparison)
overlap	O1–O5	O1, O2, O3, O4 (standard, on 48^4), O5 (standard)	O4 for 2.40 (0.405 against 0.1–0.4); the anticipated mismatch of O5 for 2.40 (not observed)
hill	H1, H2	H1, H2	—

7. Corrections

To the working notes (kept unchanged in `documenti/working_notes_it/`; the later notes and this text prevail):

- the algebra of the fifteen raw products is so(3,3), not su(4), and the sector labels N = 1 and N = 2 were swapped (4.2);
- the reading that the overlap escapes Nielsen–Ninomiya by giving up strict locality is corrected. The operator is local with exponential tails, and the decisive step is the Ginsparg–Wilson form of chiral symmetry;
- "the twin becomes non-locality" and "the twin fades over 2.5 cells" are withdrawn. The fitted slope is not the asymptotic rate (4.6);
- "the opposite hand can only be moved" is withdrawn (4.7);
- the pion ratio does not select a hand (4.6);
- the classification of E2 and E3 in the working note is replaced by section 4.5;
- the phrase "exactly the su(4) of Pati–Salam" in the Tappa 2 note is replaced by section 4.2.

To MM 2.40.

- **The valleys.** Its README (line 85) says that the massive automaton has 16 species. The full count over the Brillouin cube $[0, 2\pi)^3$ is different. Without the Wilson term ($\theta = 0.01$), the two-tick palindromic Dirac automaton has 64 Dirac valleys at distance θ from their quasi-energy centre: 32 around zero and 32 around $\pi/2$. The exact relation is $\cos \Omega = \cos 2\theta \cdot \cos 2k_x \cdot \cos 2k_y \cdot \cos 2k_z$. The two-tick evolution conserves the eight parity classes of the sites, so this is 8 valleys in each parity sector, each one a Dirac fermion. The earlier check examined 16 of them. With the Wilson term $r = 0.15$, one light species remains in the bulk (T1c).
- **The walls.** The search for wall states at the new valleys (registered T-D) found extra states at half period. The counting method used (the trace of a projector with a window of 0.08) does not isolate wall states and also catches bulk states, so D1 is reported as not met. A diagnosis made afterwards gives a first picture:
- with the 2.40 parameters ($\theta_0 = 0.2 > r = 0.15$), part of these states sit on the walls, with a mass ≈ 0.013;
- with the tested value $\theta_0 = 0.1 < r$, the selected states are predominantly bulk states.

"One mode per wall" remains valid as a local statement near $k = 0$ at quasi-energy 0. A corrected text for MM 2.40 needs a better per-eigenstate localisation criterion and will be given separately; the programs for this are not part of this deposit.

8. Reproducing

```
python codice/run_all.py            # full run
python codice/run_all.py --short    # skips the three long programs
```

Requirements: Python 3 with numpy, scipy and sympy. Reference environment: Python 3.13.13, numpy 2.4.6, scipy 1.18.0, sympy 1.14.0, Windows 11 on x86-64 (4 CPUs).

- `run_all.py` runs the included numerical programs and compares their outputs with the reference files in `risultati/`. This is a strict reproduction of the working logs, not by itself a scientific verification. It does not reproduce T-D, Tappa 3, the comparisons with published data or the bibliography checks.
- Floats must agree within $|u - v| \le 1e\text{-}9 \cdot \max(|u|, |v|) + 1e\text{-}12$; integers exactly. Quantities obtained by subtracting close numbers, and the text form "np.float64(...)" printed by numpy 2, may need the reference environment to match.
- `overlap_240.py` is stopped after its first O5 case, as in the working run; the comparison covers O1–O4 and that case.
- One line of `verifica_numeri_astra_t1.log` was appended by hand in the working records after the run. The manually appended line is excluded from the comparison; its periodicity claim is checked independently by `ricontrollo_periodo_b.py`, written for this deposit (exact for a constant mass term, as there; not extended to the k-dependent Wilson term).

- `--short` skips `tappa1_avvolgimento.py`, `overlap_240.py` and `o5_h2_rapido.py`. The Tappa 1 recheck then uses the reference `tappa1.json` as input and does not reproduce its generation.
- Before deposit the full run was made from scratch; the output and the running times are in `documenti/COLLAUDO_COMPLETO.txt`.

9. Contents

- `codice/01_premesse`: three-reading algebra (`ta_tre_letture.py`), its recheck after review (`ricontrollo_ta_dopo_astra.py`), pion ratio (`pione_elicita.py` as in the working records, `pione_elicita_v2.py` corrected).
- `codice/02_fermioni`: winding and valleys (`tappa1_avvolgimento.py`, `tappa1_dispersione.py`, `ricontrollo_dopo_tappa1.py`, `verifica_numeri_astra_t1.py`, `ricontrollo_periodo_b.py`); overlap and hill (`overlap_240.py`, `o5_h2_rapido.py`, `collina_h1.py`, with `NOTE_collina_h1.txt`: a check announced in its docstring is not executed).
- `codice/03_U1`: charges (`tappa2_carica.py`, `ricontrollo_dopo_tappa2.py`).
- `codice/04_gruppo_record`: gluons and electroweak bosons (`gluoni.py`, `ricontrollo_dopo_gluoni.py`, `debole_due_unita.py`, `ricontrollo_dopo_debole.py`).
- `risultati/`: reference outputs of the working runs.
- `documenti/REGISTERED_TESTS.md`: every registered prediction, abridged in English, with its outcome.
- `documenti/registrations_original/`: the registered files in Italian and the registration logs (`REGISTRO.txt`: UTC time, SHA-256, file name). The logs also list files of the working records that are not in this deposit.
- `documenti/working_notes_it/`: outcome notes in Italian, including the notes written after each review.
- `WHERE_THIS_COMES_FROM.md`: optional background.
- `LEGGIMI_IT.md`: summary in Italian. `IMPRONTE.sha256`: fingerprints of every file.

10. How this was made

The author is not a professional physicist. The premises are his. The translations into physical terms, the programs and this text were made with an AI assistant (Claude), and are marked as the translator's where they add to the premises. The predictions were fingerprinted before computing in the author's working files; these fingerprints are not publicly time-stamped. A second AI system (gpt-6-astra via Codex) reviewed each working round and this package, and recomputed several numbers independently. Its corrections are applied (section 7). Its new numbers were rechecked with code (`verifica_numeri_astra_t1.py`, `ricontrollo_ta_dopo_astra.py`, and a root search for the complex-momentum singularities of section 4.6). No human peer review is claimed. The literature check covers the works listed in section 2, not the whole literature.

Glossary

- **Reading**: in MM, a registered observation; here, the way a field is acted on by a force (which components, which hand).
- **Link transport**: the matrix that carries the internal components of a field from one lattice site to the next (a phase for light, 2×2 for the weak force, 3×3 for colour).
- **Hand**: chirality, left or right.
- **Twin, doubler**: an extra light species forced by the lattice (Nielsen–Ninomiya).
- **Hill**: the Wilson term r Σ(1 − cos k_μ), zero at k = 0 (the valley) with ridges 2rn at the corners with n components equal to π.
- **Reading level m_0**: the mass parameter of the overlap; the light species are those with Wilson term below m_0.
- **Winding W**: the three-dimensional winding number of a map from the Brillouin torus to SU(2) or U(n). For SU(2) ≅ S^3 it is the degree of the map; when ±I are regular values it equals the sum of the node chiralities, defined from the local effective Hamiltonian, at quasi-energy 0, and separately at π.
- **Registered**: written to a file and fingerprinted before the calculation.

References

Works cited in the record, with their DOI.

- A et al. (2000). Hierarchies without symmetries from extra dimensions. https://doi.org/10.1103/physrevd.61.033005
- Bagdonaite et al. (2013). Robust Constraint on a Drifting Proton-to-Electron Mass Ratio at z=0.89 from Methanol Observation at Three Radio Telescopes. https://doi.org/10.1103/physrevlett.111.231101
- Bisio et al. (2014). Quantum field as a quantum cellular automaton: The Dirac free evolution in one dimension.

https://doi.org/10.1016/j.aop.2014.12.016
- Burns et al. (2024). Constraints on variation of the weak scale from big bang nucleosynthesis. https://doi.org/10.1103/physrevd.109.123506
- D’Ariano et al. (2014). Derivation of the Dirac equation from principles of information processing. https://doi.org/10.1103/physreva.90.062106
- Furey (2014). Generations: three prints, in colour. https://doi.org/10.1007/jhep10(2014)046
- Furey (2016). Charge quantization from a number operator. https://doi.org/10.48550/arxiv.1603.04078
- Ginsparg et al. (1982). A remnant of chiral symmetry on the lattice. https://doi.org/10.1103/physrevd.25.2649
- Higashikawa et al. (2019). Floquet Chiral Magnetic Effect. https://doi.org/10.1103/physrevlett.123.066403
- Kadoh et al. (2007). A construction of the Glashow-Weinberg-Salam model on the lattice with exact gauge invariance. https://doi.org/10.48550/arxiv.0709.3658
- Kaplan (2011). Chiral symmetry and lattice fermions. https://doi.org/10.1093/acprof:oso/9780199691609.003.0004
- Lüscher (2002). Chiral gauge theories revisited. https://doi.org/10.1142/9789812778253_0002
- Navas et al. (2024). Review of Particle Physics. https://doi.org/10.1103/physrevd.110.030001
- Nielsen et al. (1981). Absence of neutrinos on a lattice. https://doi.org/10.1016/0550-3213(81)90361-8
- Pati et al. (1974). Lepton number as the fourth "color". https://doi.org/10.1103/physrevd.10.275
- Shaposhnikov et al. (2009). Asymptotic safety of gravity and the Higgs boson mass. https://doi.org/10.1016/j.physletb.2009.12.022
- Sun et al. (2018). Three-Dimensional Chiral Lattice Fermion in Floquet Systems. https://doi.org/10.1103/physrevlett.121.196401
- Uzan (2011). Varying Constants, Gravitation and Cosmology. https://doi.org/10.12942/lrr-2011-2
- Wilson (1974). Confinement of quarks. https://doi.org/10.1103/physrevd.10.2445
- Yoo et al. (2003). Big bang nucleosynthesis and cosmic microwave background constraints on the time variation of the Higgs vacuum expectation value. https://doi.org/10.1103/physrevd.67.043517